

PRE-SELECTION TEST 2026-27
CLASS X – ROBOTICS AND ARTIFICIAL INTELLIGENCE

Maximum Marks: 50

Time Allowed: One Hour

Instructions:

- Attempt all questions.
- Marks are indicated in brackets next to each question.

Section A

(Attempt all questions)

[1 × 10 = 10 marks]

Question 1: Write the correct answer only. For example: [I.(a) Answer]

- I. Which of the following is an example of a New Age Robotic System (NARS)?
(a) A basic desk calculator (b) An assistant robot used in hospital patient care (c) A simple wall clock (d) A mechanical timer
- II. Cobots work:
(a) Independently (b) With other machines (c) With humans (d) Only in schools
- III. Which type of actuator produces motion along a straight line?
(a) Rotary actuator (b) Linear actuator (c) Vision sensor (d) Controller
- IV. Which built-in Python function returns the number of characters in a string?
(a) `length()` (b) `len()` (c) `size()` (d) `count()`
- V. Camera is an example of a:
(a) Vision Sensor (b) Pressure Sensor (c) Motion Sensor (d) Solar Sensor
- VI. Which escape sequence is used to insert a newline in a Python string?
(a) `\t` (b) `\n` (c) `\r` (d) `\s`
- VII. Manual control differs from automatic control mainly because manual control requires:
(a) Sensors only (b) Human intervention (c) Actuators only (d) A closed feedback loop
- VIII. Drones are primarily used for:
(a) Underwater exploration only (b) Aerial tasks such as surveillance and delivery (c) Only military purposes (d) Long-term data storage
- IX. Which of the following is a key advantage of cobots over traditional industrial robots?
(a) They work best in isolation (b) They can work safely alongside humans (c) They require no safety measures at all (d) They cannot be reprogrammed
- X. Which Python list method is used to add a single element to the end of a list?
(a) `add()` (b) `append()` (c) `insert_new()` (d) `push()`

Question 2: Answer the following in brief.

[2 × 5 = 10 marks]

- a) Write two examples of New Age Robotic Systems (NARS) used in everyday life (in detail). [2]
- b) Write FOUR limitations of cobots. [2]
- c) Concatenate these strings "Robotics" and "AI" with a dash(-) in between using Python. [2]
- d) Write the output of the following code with **explanation**. [2]

```
1 text = "Python Programming"
2 print(text[2:11:2])
```

- e) Can strings be concatenated with integers directly? Explain with a example. [2]

Section B

(Attempt all questions)

[15 × 2 = 30 marks]

Question 3

- a) How do warehouse robots assist in logistics and inventory management? Explain with examples. [3]
- b) What is a control system? Explain these components of a control system: **Set Point/reference**, **Controller**, **Actuator**, **Plant**, and **Feedback**. Draw the block diagram [1+3+1]
- c) Why new age robotic systems are relevant? (Four points) [2]
- d) State and describe the function of any two sensors classified by application (for example: proximity, temperature, tactile, or range sensors). [3]
- e) Write the output of the following code with **explanation**. [2]

```
1 print("Hello".isalpha())
2 print("abc123".isalnum())
```

Question 4

- a) Write the principles of gears. If a 30-tooth driver gear rotates at 300 RPM, how many teeth does the driven gear have if it rotates at 450 RPM? [2+3]
- b) What is the importance of an actuator? Write two main differences between rotary actuators and linear actuators. [1+2]
- c) Write differences between manual vs automatic control systems in terms of speed, accuracy and adaptability. [2]
- d) Write a Python program that: [5]
 - Accepts a list of student marks as input
 - Appends one new mark to the list
 - Sorts the list in ascending order
 - Prints the highest and lowest marks in the final list